Intro to
Optimization Under Uncertainty

2026 Fall
AD7031

AI Changes World

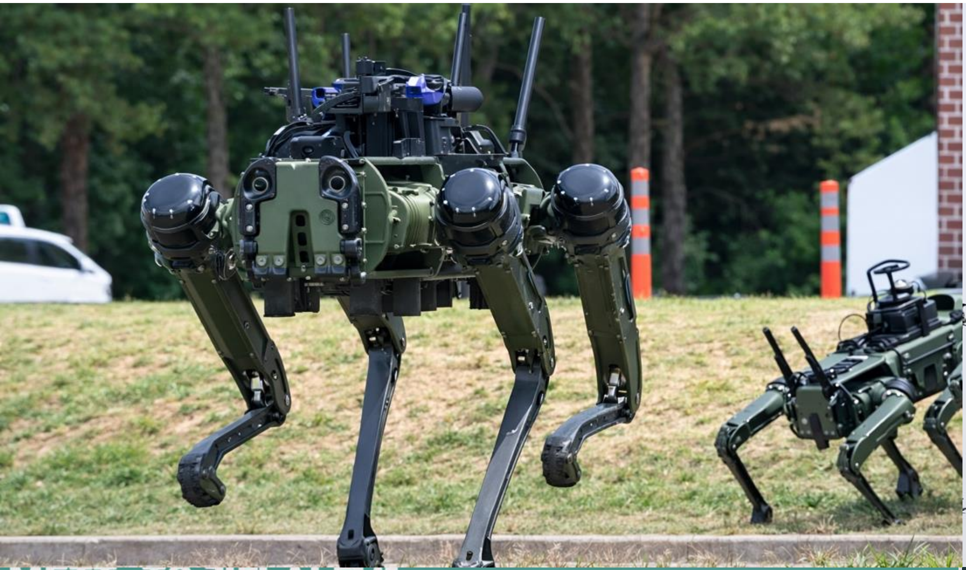
ChatGPT

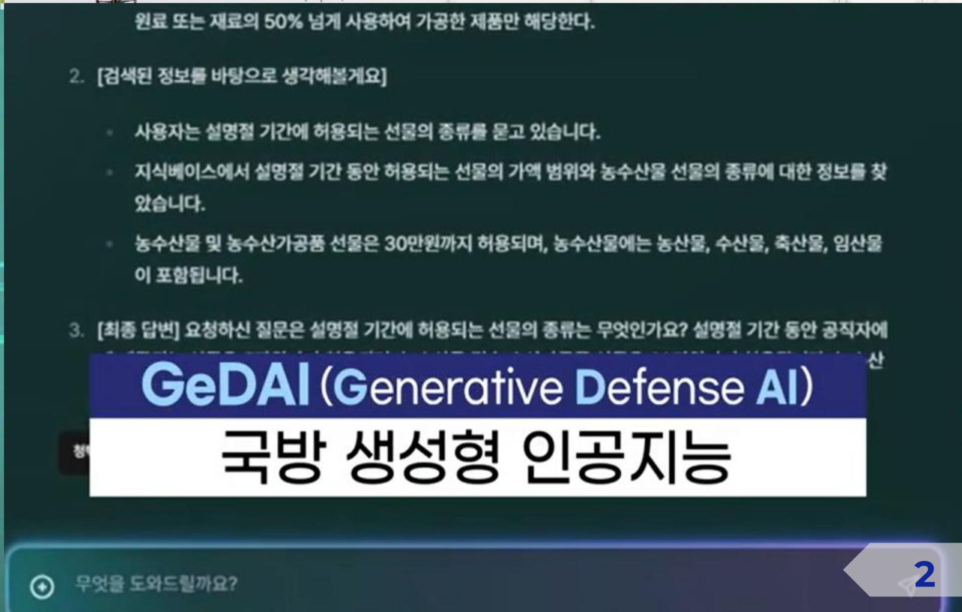
Gemini

perplexity

Claude

AI in Defense





Why Robust AI?

Boston Dynamics Atlas

Tesla Autopilot

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MIT Technology Review

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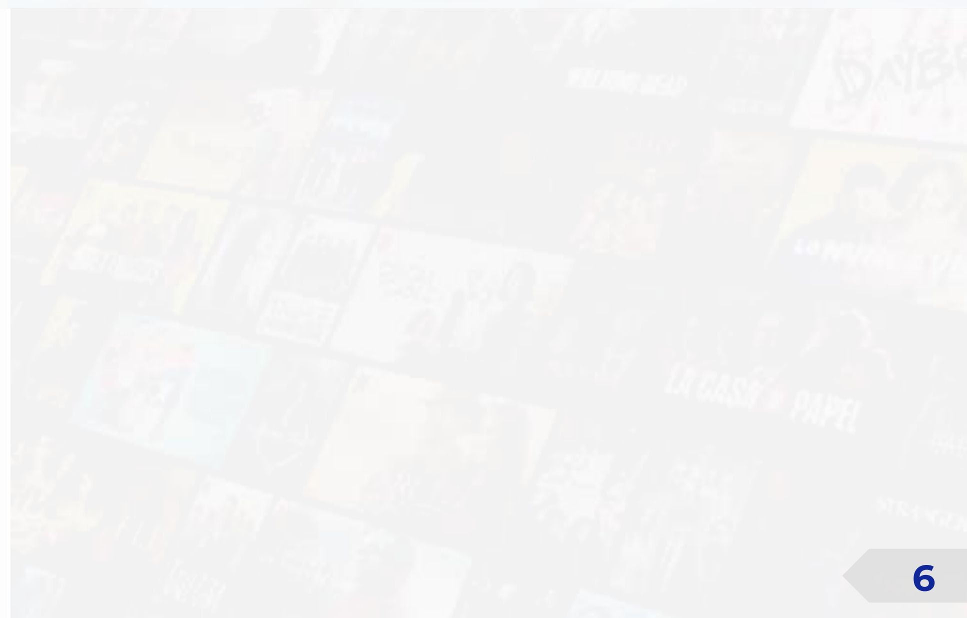
ARTIFICIAL INTELLIGENCE

The way we train AI is fundamentally flawed

The process used to build most of the machine-learning models we use today can't tell if they will work in the real world or not—and that's a problem.

AI is not magic —

it is optimization under uncertainty



Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{\boldsymbol{x}} \\ \tilde{\boldsymbol{y}} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then

the best model parameter $\boldsymbol{\theta}^\star$ is

$$\boldsymbol{\theta}^\star \in \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{\mathbb{P}}\left[\ell\left(\tilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}})\right)\right]$$

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Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Regression : predict continuous $\boldsymbol{y} \in \mathbb{R}^n$
- Classification : predict binary $\boldsymbol{y} \in \{-1, +1\}$

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- Regression : predict continuous $\boldsymbol{y} \in \mathbb{R}^n$

$$\textbf{Squared Loss}(\cdot) = \|\cdot\|_2^2 = \|\boldsymbol{y} - f_{\boldsymbol{\theta}}(\boldsymbol{x})\|_2^2$$

e.g., when n=2

$$\boldsymbol{y}_1 = \begin{bmatrix} 100 \\ 10 \end{bmatrix}, f_{\boldsymbol{\theta}}(\boldsymbol{x}_1) = \begin{bmatrix} 105 \\ 9 \end{bmatrix}$$

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$$\boldsymbol{y}_1 = \begin{bmatrix} 100 \\ 10 \end{bmatrix}, f_{\boldsymbol{\theta}}(\boldsymbol{x}_1) = \begin{bmatrix} 105 \\ 9 \end{bmatrix}$$

$$\Rightarrow \left\|\begin{bmatrix} 100 \\ 10 \end{bmatrix} - \begin{bmatrix} 105 \\ 9 \end{bmatrix}\right\|_2^2 = \left\|\begin{bmatrix} -5 \\ 1 \end{bmatrix}\right\|_2^2 = (-5)^2 + 1^2 = 26$$

Optimization under Uncertainty: ML

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e.g., when n=1

$$y_1 = 10, f_{\boldsymbol{\theta}}(x_1) = 11$$

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e.g., when n=1

$$y_1 = 10, f_{\boldsymbol{\theta}}(\boldsymbol{x}_1) = 11$$

$$\Rightarrow \|10 - 11\|_2^2 = \|-1\|_2^2 = (-1)^2 = 1$$

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- Classification : predict binary $y \in \{-1, +1\}$

$$\textbf{0–1 Loss}(\cdot) = \begin{cases} 0, & \text{if } \mathrm{sign}(f_{\boldsymbol{\theta}}(\boldsymbol{x})) = \mathrm{sign}(\boldsymbol{y}) \\ 1, & \text{if } \mathrm{sign}(f_{\boldsymbol{\theta}}(\boldsymbol{x})) \neq \mathrm{sign}(\boldsymbol{y}) \end{cases}$$

e.g., $\boldsymbol{y}_1 = -1, f_{\boldsymbol{\theta}}(\boldsymbol{x}_1) = -1000.5$

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e.g., $\boldsymbol{y}_1 = -1, f_{\boldsymbol{\theta}}(\boldsymbol{x}_1) = -1000.5 \Rightarrow$ Correct! 0-loss

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e.g., $\boldsymbol{y}_1 = -1, f_{\boldsymbol{\theta}}(\boldsymbol{x}_1) = -1000.5 \Rightarrow$ Correct! 0-loss

$\boldsymbol{y}_2 = +1, f_{\boldsymbol{\theta}}(\boldsymbol{x}_2) = -1$

Optimization under Uncertainty: ML

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e.g., $y_1 = -1, f_{\boldsymbol{\theta}}(x_1) = -1000.5 \Rightarrow$ Correct! 0-loss

$y_2 = +1, f_{\boldsymbol{\theta}}(x_2) = -1 \Rightarrow$ Wrong! 1-loss

Optimization under Uncertainty: ML

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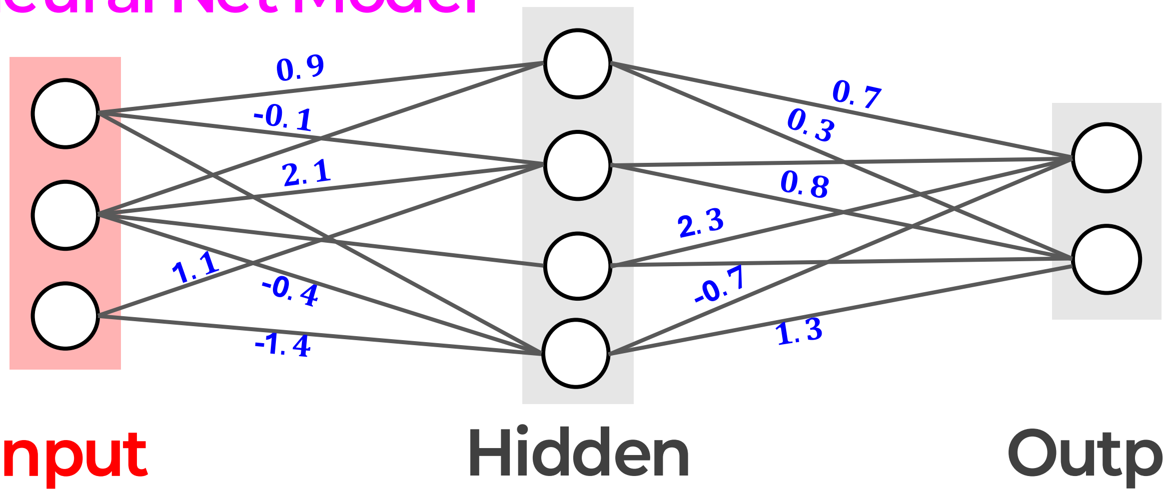
Linear Model

$$f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}}) = \boldsymbol{\theta}^\top \boldsymbol{x} = \sum_i x_i \theta_i$$

Neural Net Model



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ML $\subset$ optimization under uncertainty

- OR :

$$\min_{\boldsymbol{x}} \mathbb{E}_{\mathbb{P}}\left[f\left(\boldsymbol{x}, \tilde{\xi}\right)\right]$$

$$\text{s.t} \quad \text{Prob}\left[g\left(\boldsymbol{x}, \tilde{\xi}\right) \leq 0\right] \geq \alpha$$

- Control

$$\min_{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{T-1}} \mathbb{E}_{\mathbb{P}}\left[\sum_{t=1}^{T-1} c_t\left(\tilde{\boldsymbol{x}}_t, \boldsymbol{u}_t, \tilde{\xi}_t\right) + c_T\left(\tilde{\boldsymbol{x}}_T\right)\right]$$

$$\text{s.t} \quad \tilde{\boldsymbol{x}}_{t+1} = f_t\left(\tilde{\boldsymbol{x}}_t, \boldsymbol{u}_t, \tilde{\xi}_t\right) \; \forall t = 1, \ldots, T-1$$

Optimization under Uncertainty: ML

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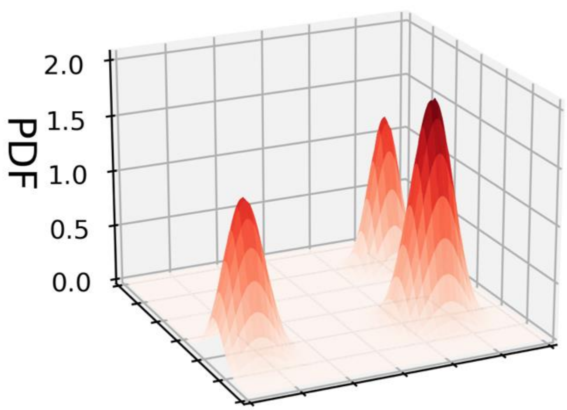
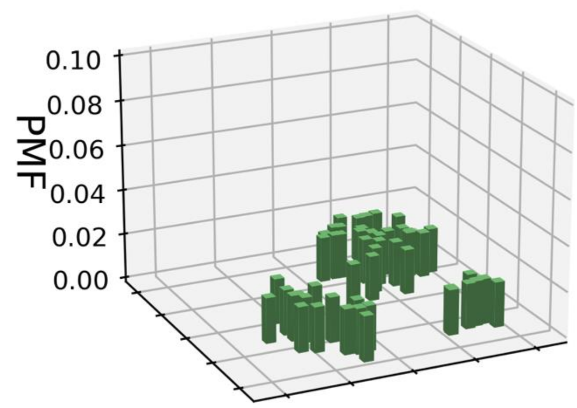
$$\boldsymbol{\theta}^\star \in \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{\mathbb{P}}\left[\ell\left(\tilde{y}, f_{\boldsymbol{\theta}}(\tilde{x})\right)\right]$$

$\mathbb{P}$ unknown $\Longrightarrow$ $\boldsymbol{\theta}^\star$ not available

Only N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ available: $\left(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}\right)$



True distribution $\mathbb{P}$



Empirical distribution $\widehat{\mathbb{P}}_N$

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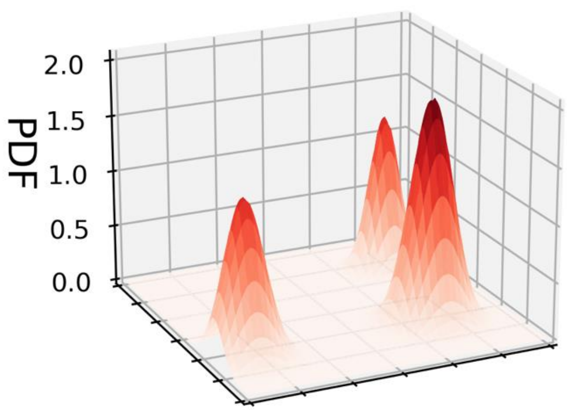

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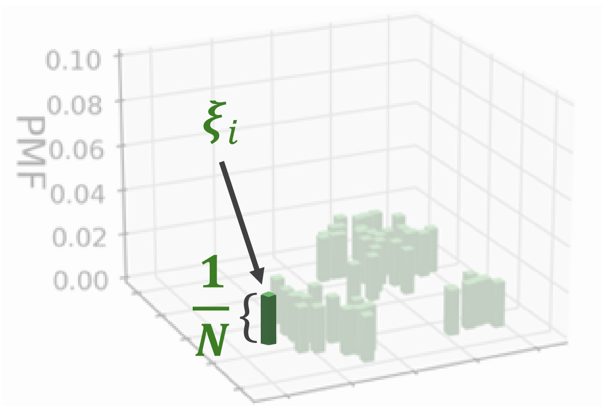

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Data-Driven Optimization

Assume $\xi \sim$empirical distribution $\widehat{\mathbb{P}}_N$, then the best empirical parameter $\widehat{\boldsymbol{\theta}}_N$ is

$$\begin{aligned} \widehat{\boldsymbol{\theta}}_N &\in \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{\widehat{\mathbb{P}}_N}\left[\ell\left(\tilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}})\right)\right] \\ &= \arg\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^{N} \ell(\boldsymbol{y}_i, f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)) \end{aligned}$$

Optimization under Uncertainty: ML

$$\boldsymbol{\theta}^\star \in \overbrace{\operatorname*{arg\,min}_{\boldsymbol{\theta}} \mathbb{E}_{\mathbb{P}}\left[\ell\left(\tilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}})\right)\right]}^{\text{True Problem}}$$

$$\neq$$

$$\widehat{\boldsymbol{\theta}}_N \in \underbrace{\operatorname*{arg\,min}_{\boldsymbol{\theta}} \mathbb{E}_{\widehat{\mathbb{P}}_N}\left[\ell\left(\tilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}})\right)\right]}_{\text{Data-driven Problem}}$$

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If $\widehat{\mathbb{P}}_N \approx \mathbb{P}$, then $\widehat{\boldsymbol{\theta}}_N$ is a good approximation to $\boldsymbol{\theta}^\star$

- In ML , $\widehat{\boldsymbol{\theta}}_N$ is called Empirical Risk Minimizer (ERM)

- In OR , it is called Sample Average Approximation (SAA)

Optimization under Uncertainty: ML

$$\boldsymbol{\theta}^\star \neq \widehat{\boldsymbol{\theta}}_N\,!$$

If $\widehat{\mathbb{P}}_N$ poorly approximates $\mathbb{P}$—especially with sparse data—

$\widehat{\boldsymbol{\theta}}_N$ may perform poorly.

Optimization under Uncertainty: ML

$$\boldsymbol{\theta}^\star \in \overbrace{\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{\mathbb{P}}\left[\ell\left(\tilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}})\right)\right]}^{\text{True Problem}}$$

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$$\widehat{\boldsymbol{\theta}}_N \in \underbrace{\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{\widehat{\mathbb{P}}_N}\left[\ell\left(\tilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}})\right)\right]}_{\text{Data-driven Problem}}$$

If $\widehat{\mathbb{P}}_N$ poorly approximates $\mathbb{P}$—especially with sparse data—

$\widehat{\boldsymbol{\theta}}_N$ may perform poorly.

Distributionally Robust Optimization (DRO)

$$\min_{\boldsymbol{\theta}} \mathbb{E}_{\widehat{\mathbb{P}}_N}\left[\ell\left(\widetilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\widetilde{\boldsymbol{x}})\right)\right]$$

Distributionally Robust Optimization (DRO)

$$\min_{\boldsymbol{\theta}} \max_{\mathbb{Q} \in \mathcal{P}_\lambda(\widehat{\mathbb{P}}_N)} \mathbb{E} \left[\ell\left(\tilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}})\right) \right]$$

Distributionally Robust Optimization (DRO)

$$\min_{\boldsymbol{\theta}} \max_{\mathbb{Q} \in \mathcal{P}_{\lambda}(\widehat{\mathbb{P}}_N)} \mathbb{E}_{\mathbb{Q}}\left[\ell\left(\tilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}})\right)\right]$$

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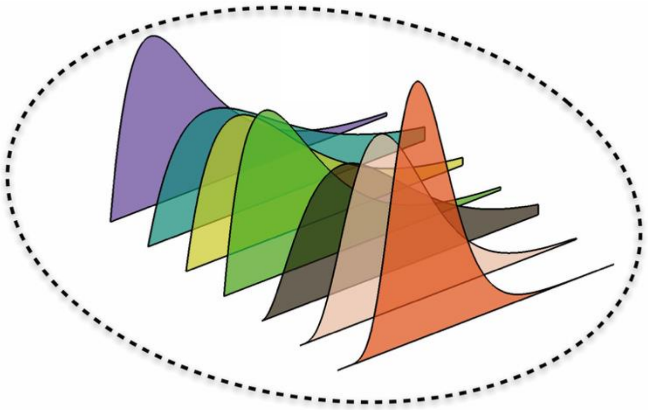
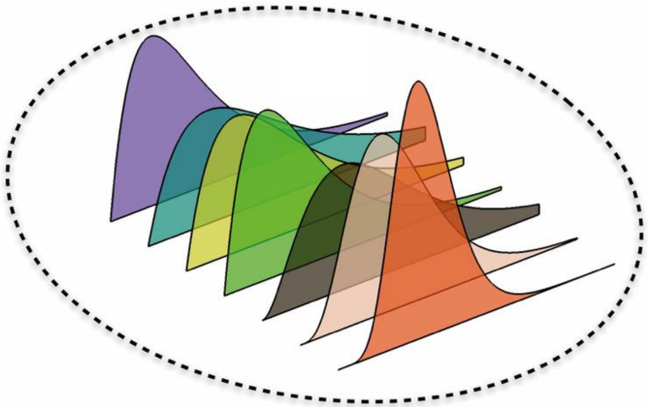


$\mathcal{P}_\lambda(\cdot)$: Ambiguity set

Distributionally Robust Optimization (DRO)

$$\widehat{\boldsymbol{\theta}}_N^{DRO} \in \arg\min_{\boldsymbol{\theta}} \max_{\mathbb{Q} \in \mathcal{P}_\lambda(\widehat{\mathbb{P}}_N)} \mathbb{E}_{\mathbb{Q}}\left[\ell\left(\tilde{\boldsymbol{y}}, f_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}})\right)\right]$$

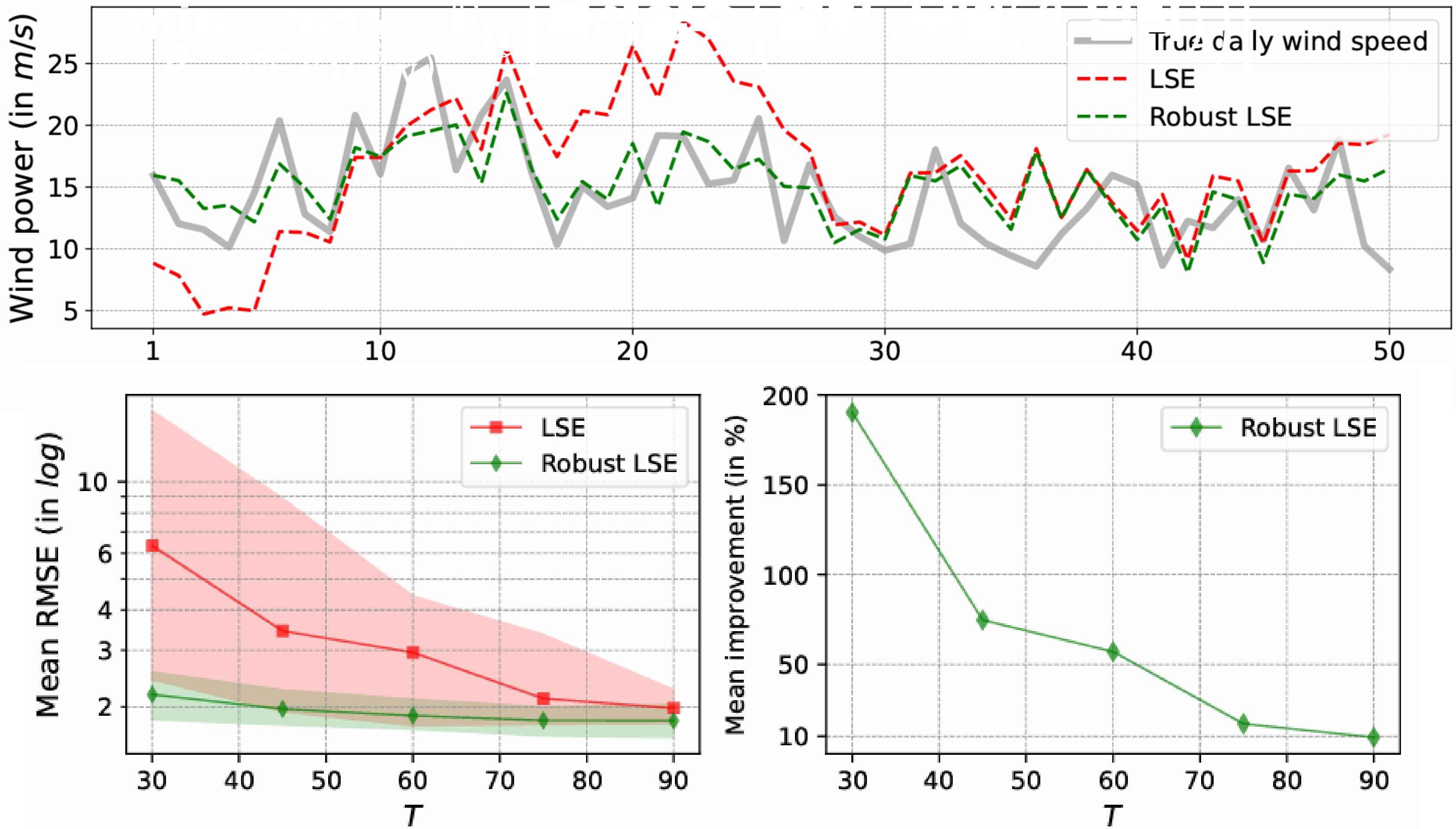


$\mathcal{P}_\lambda(\cdot)$: Ambiguity set

$\widehat{\boldsymbol{\theta}}_N^{DRO} \Rightarrow$ Robust performance under data scarcity

Machine Learning

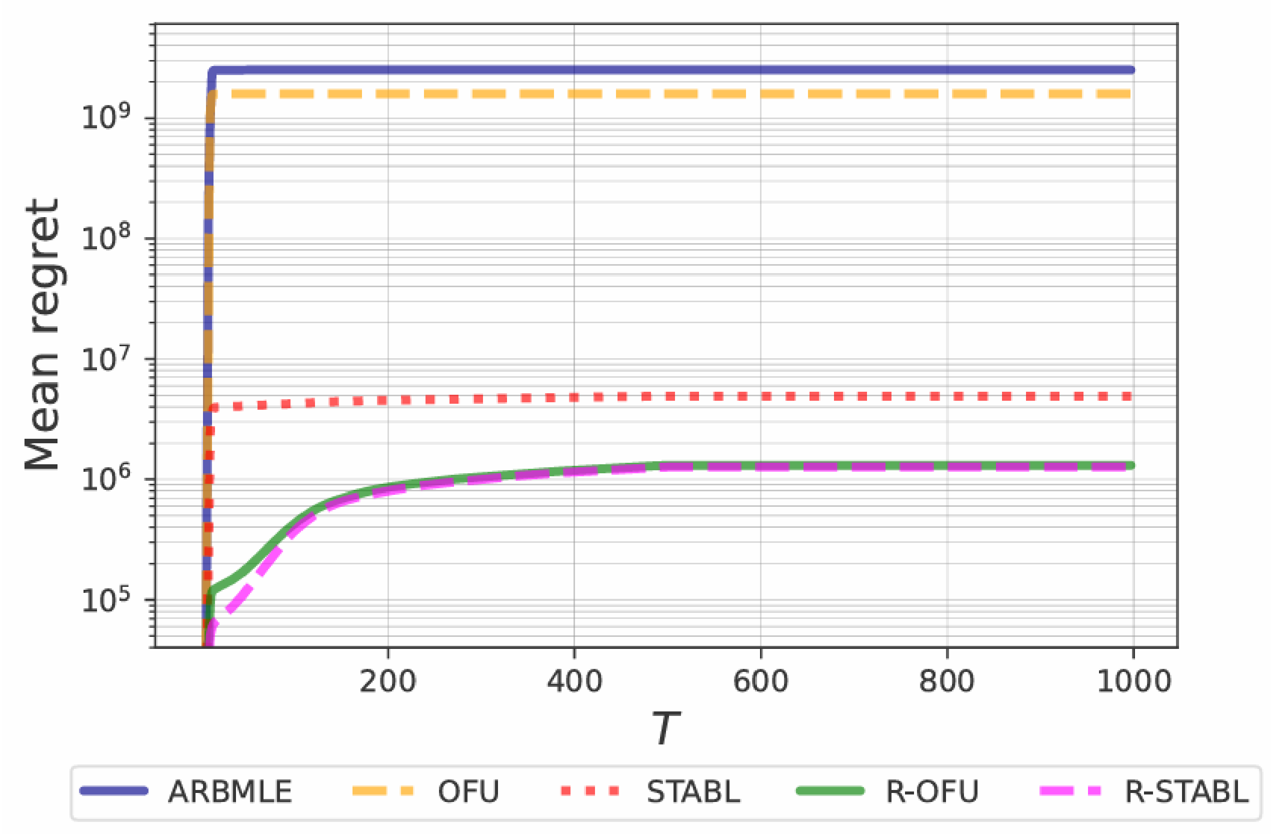
DRO-based NN model : Short-term Wind Prediction [4)]



4) Park et al., "Robust System Identification", *ICLR*, 2025

Machine Learning

DRO-based Reinforcement Learning : Boeing 747 Control [4)]



4) Park et al., "Robust System Identification", *ICLR*, 2025

Operations Research

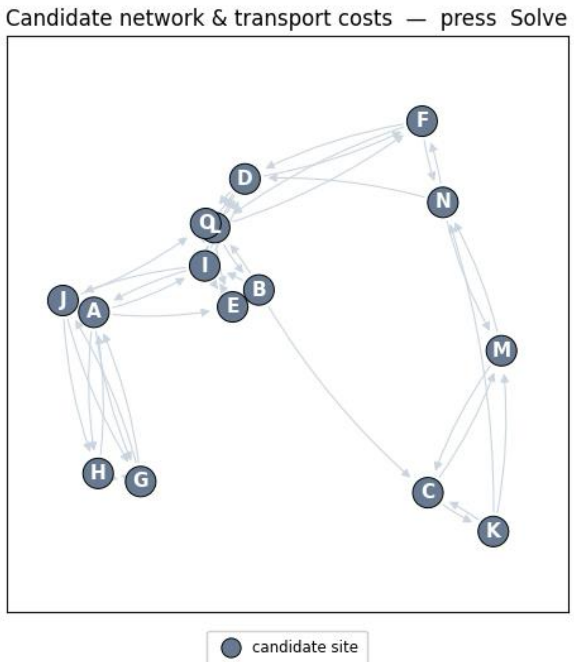
DRO SDDP : Portfolio Optimization [5)]

Dataset	Scheme	Utility	Mean return	Std. Dev.	Sharpe ratio
10 Industry Fama-French	DDR-SDDP	3.19E-02	3.41E-02	6.35E-02	4.77E-01
	DD-SDDP	2.65E-02	2.90E-02	6.95E-02	3.62E-01
	Independent	1.58E-02	1.76E-02	6.23E-02	2.21E-01
	HMM-SDDP	2.43E-02	2.54E-02	4.70E-02	4.59E-01
	Equal	-2.15E-03	-1.97E-03	2.74E-02	-3.52E-02
12 Industry Fama-French	DDR-SDDP	2.49E-02	2.67E-02	5.97E-02	3.82E-01
	DD-SDDP	2.35E-02	2.59E-02	6.90E-02	3.20E-01
	Independent	2.01E-02	2.20E-02	6.32E-02	2.88E-01
	HMM-SDDP	1.97E-02	2.09E-02	4.97E-02	3.42E-01
	Equal	-1.52E-03	-1.36E-03	2.72E-02	-3.17E-02
12 Industry Fama-French	DDR-SDDP	1.38E-02	1.43E-02	3.61E-02	2.89E-01
	DD-SDDP	7.34E-03	9.45E-03	6.86E-02	8.16E-02
	Independent	3.43E-03	5.34E-03	6.51E-02	2.28E-02
	HMM-SDDP	5.68E-03	6.09E-03	3.43E-02	6.53E-02
	Equal	-5.00E-02	-4.87E-02	2.73E-02	-3.18E-01

5) Park et al., "Data-driven Stochastic Dual Dynamic Programming", *SIOPT*, 2026

Operations Research

DRO Two-stage LAP (Location-Allocation Problem) : Drone Scheduling



Two-stage LAP (Sample Average Approximation)

Stage 1 (here-and-now, before the disaster)

$$\min_{\mathbf{x},\mathbf{r}} \sum_j c_j x_j + \sum_j f_j r_j + \frac{1}{N}\sum_{n=1}^{N} Q(\mathbf{x},\mathbf{r};\hat{\boldsymbol{\xi}}^n)$$

$$\text{s.t. } \sum_{j\in\mathcal{N}} r_j \le B, \quad \sum_{j\in\mathcal{N}} x_j \le p$$

$$r_j \le Bx_j, x_j \in \{0,1\}, r_j \ge 0 \quad \forall j\in\mathcal{N}$$

Stage 2 (recourse for scenario n, after demand is seen)

$$Q(\mathbf{x},\mathbf{r};\hat{\boldsymbol{\xi}}^n) = \min_{\mathbf{y},\mathbf{q},\mathbf{s}} \sum_a d_a y_a + \sum_j u_j q_j + \sum_j o_j s_j$$

$$\text{s.t. } (\text{in}_j - \text{out}_j) + q_j - s_j = \hat{\xi}_j^n - r_j \quad \forall j\in\mathcal{N}$$

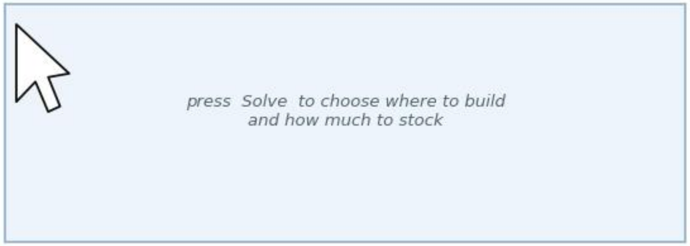
$$0 \le y_a \le \bar{y} \ \forall a\in\mathcal{A}; \quad 0 \le q_j \le \hat{\xi}_j^n, s_j \ge 0$$

DECISION VARIABLES (what we choose)
s1: $\mathbf{x}$ open facilities, $\mathbf{r}$ reserves; s2: $\mathbf{y}$ flows, $\mathbf{q}$ shortages, $\mathbf{s}$ surpluses

UNCERTAINTY (seen only after stage 1)
$\hat{\boldsymbol{\xi}}^n$ = demand vector in scenario n ($\hat{\xi}_j^n$ at node j)

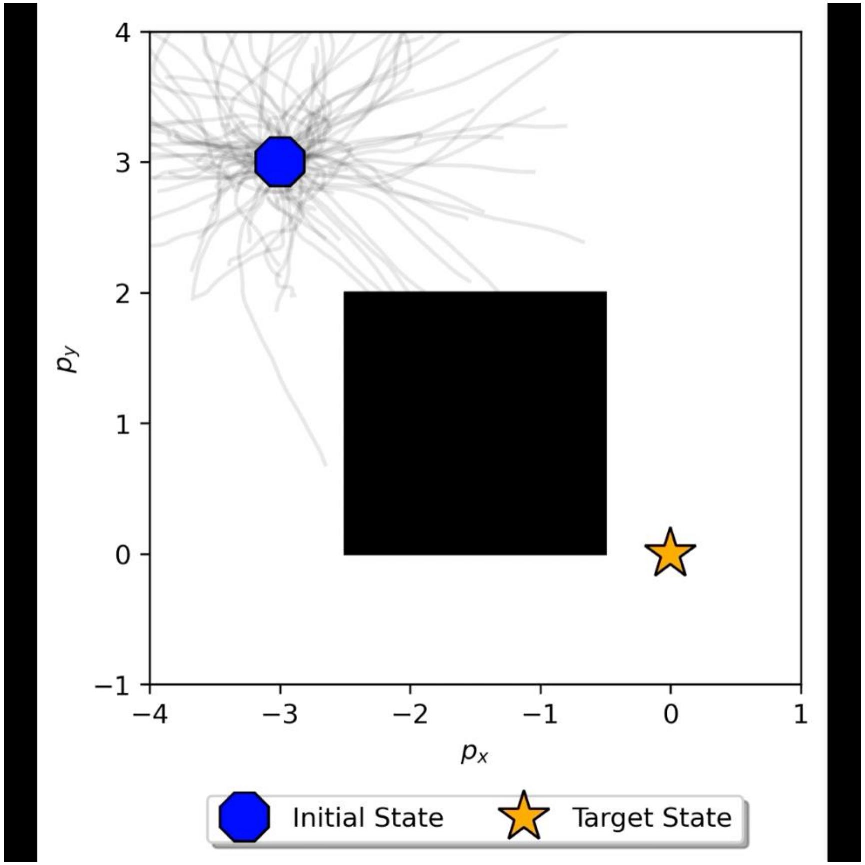
PARAMETERS (given numbers)
c_j open, f_j reserve, B budget, p cap; d_a transport, u_j short, o_j surplus, N #scen

SETS & INDICES
graph $G = (\mathcal{N},\mathcal{A})$: $\mathcal{N}$ nodes, $\mathcal{A}$ arcs; $j\in\mathcal{N}$, $a\in\mathcal{A}$, $n = 1, \dots, N$



Control

DRO PIC (Path Integral Control) : Robotics [6)]

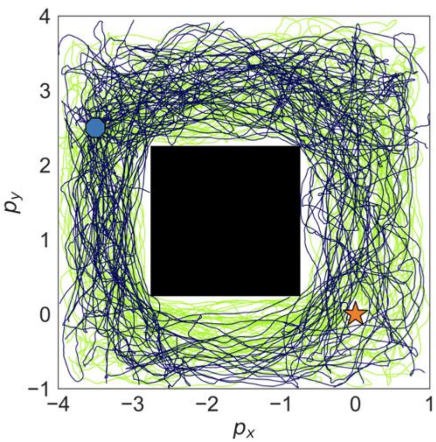


6) Park et al., "Distributionally Robust Path Integral Control," *Proc. ACC*, 2025.

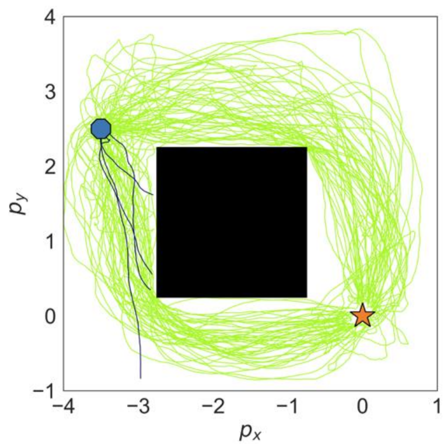
Control

DRO PIC (Path Integral Control) : Robotics 6)

- Particle robot

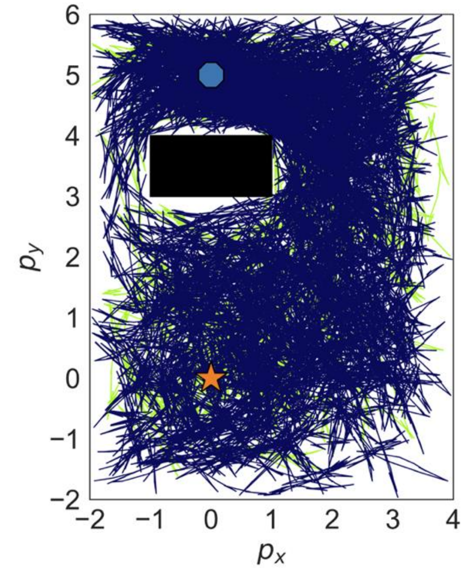


RN-PIC

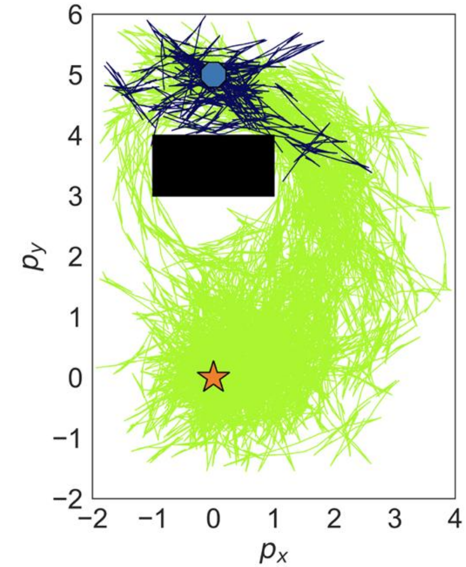


DR-PIC

- Unicycle



RN-PIC

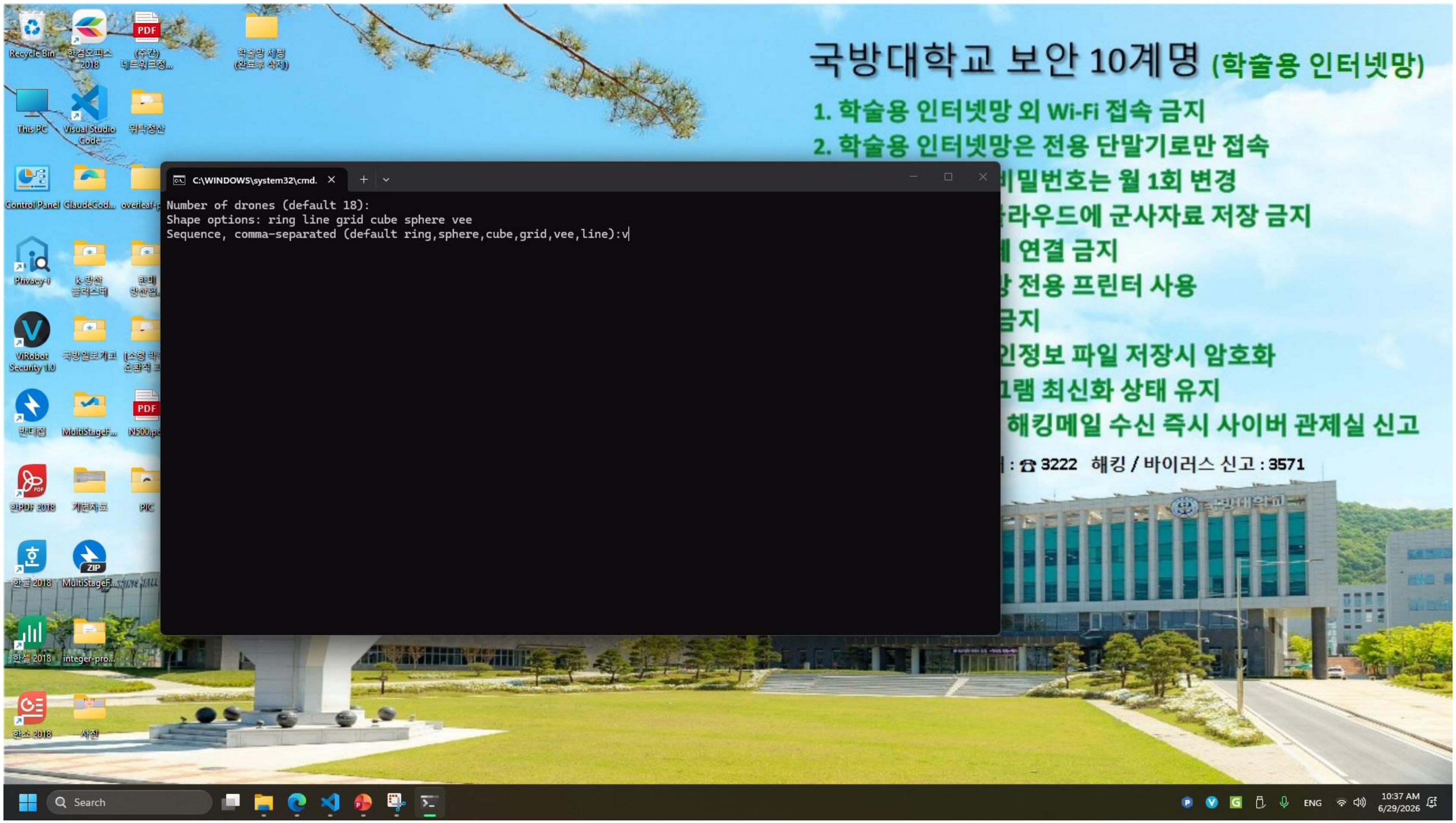


DR-PIC

6) Park et al., "Distributionally Robust Path Integral Control," *Proc. ACC*, 2025.

Control

DRO PIC (Path Integral Control) : Robotics [6)]



6) Park et al., "Distributionally Robust Path Integral Control," *Proc. ACC*, 2025.

Thanks!
Any question?